Skincare Recommendation

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The Data

We focused on two main datasets sourced from Kaggle: “Skincare Products and their Ingredients” and “Skincare Products Sentiment Analysis”

Skin Care Products

- Most important columns: product_id, ingredients, highlights, brand_name, primary_category, rating, reviews
- Help us identify and sort the products, while also seeing if the product is popular or not

Skincare Product Reviews

- Most important columns: product_id, rating, is_recommended, review_text, user_attributes
- Linking products to their ratings and reviews, while seeing how they are connected to an individual's’ unique attributes

Data Cleaning and EDA: Part 1

Initially, both the datasets had unnecessary columns that either held no values or included additional information that we did not require for our model.

Skin Care Products

- What we removed:
  - Redundant pricing columns (value_price_usd, sale_price_usd)
  - "Variation_desc"
  - "Value and Gift Sets"
- What the missing values looked like:
  - Missing values in rating and reviews
    - Replacing missing values with the median rating

Skincare Product Reviews:

- What we removed:
  - Unnamed columns
  - "Review_title, helpfulness"
  - Rows with missing user attributes
- What the missing values looked like:
  - Missing values in user_attributes:
    - Skin_type, skin_tone, eye_color, hair_color
  - Removed low information rows for a higher quality model

Data Cleaning and EDA: Part 2

After removing columns with excessive missing values, we filtered dataset to include only skincare products, dropped rows where values in most important columns are missing and then filled in remaining missing values.

Example:

1)

Before

secondary_category	tertiary_category
Cleansers	N/A

After

secondary_category	tertiary_category
Cleansers	Cleansers

2)

Before

product	rating	reviews
A	4.5	120
B	N/A	85
C	4.8	N/A

After (Median Imputation)

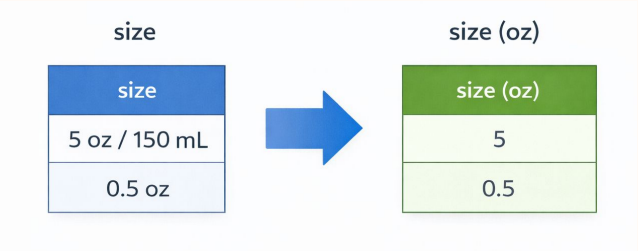
product	rating	reviews
A	4.5	120
B	4.6*	85
C	4.8	95*

* Filled with median value

Data Cleaning and EDA: Part 3

Feature Engineering

1) Converted text-based “size” descriptions into numeric values



2) Converted 10+ categorical skin tones into 3 binary features

3) Converted categorical skin types into binary features

Text Mining:

Part 1

Columns we texted mined:

Skincare Dataset:
- 'ingredients'
- 'highlights'

Review Dataset:
- 'review_text'

Ingredients

Step 1 : Turning Ingredients String Into Long List

Cleaned formatting and tokenized by commas.

Raw Ingredients	Parsed Ingredients
['C12-15 Alkyl Benzoate, Ethylhexyl Palmitate, Caprylic/Capric Triglyceride, Glycerin, Water (Aqua), Sucrose Laurate, Olea Europaea (Olive) Fruit Oil,...	c12-15 alkyl benzoate, ethylhexyl palmitate, caprylic/capric triglyceride, glycerin, water, sucrose laurate...

Step 2: Identify Categories

Grouped by skincare functions:

	Ingredient Categories
0	Cleansers
1	Moisturizers
2	Exfoliants
3	Actives
4	Soothing
5	Preservatives
6	Fragrance
7	Oils
8	Water Based

Step 3: Create Count + Binary Columns

	product_name	count_cleansers	count_moisturizers	count_exfoliants	count_actives	count_soothing	count_preservatives	count_fragrance	count_oils	count_water_based
15	GENIUS Ultimate Anti-Aging Melting Cleanser	0	1	0	0	0	3	2	6	1
16	Gentle Rejuvenating Cleanser	3	2	3	0	0	2	2	0	2
17	Advanced Anti-Aging Repairing Oil	0	2	0	1	0	0	0	4	0
18	ALIVE Prebiotic Balancing Mask	5	2	4	1	0	3	1	0	2
19	Balancing Cleanser	1	1	1	0	1	2	2	0	1

Highlights

	Highlight Categories
0	Acne
1	Aging
2	Pores
3	Redness
4	Dark Spots
5	Texture
6	Hydrating
7	Brightening
8	Soothing
9	Plumping
10	Firming

Grouped review terms to skincare concerns and benefits

Count:
How many ingredients in that category.

Binary:
Whether it exists or not.

Text Mining:

Part 2
REVIEWS

EDA on reviews using TfidfVectorizer

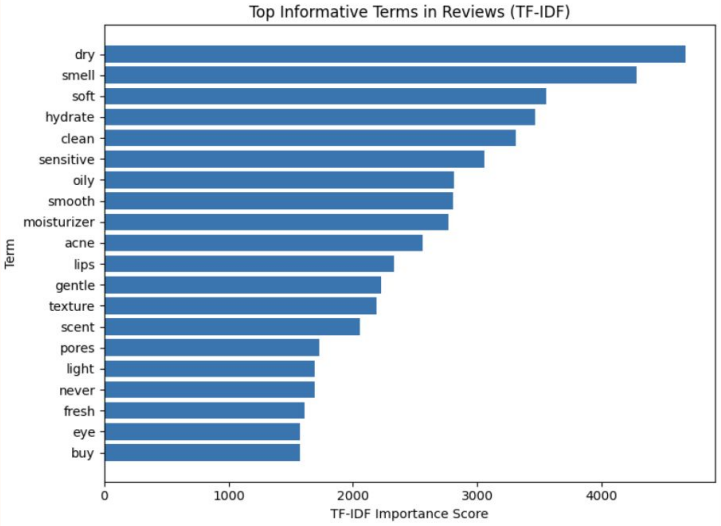
- Identified top unigrams & bigrams
- Extracted dominant skincare themes

UNIGRAMS

	term	score
0	dry	4677.232307
1	smell	4279.687857
2	soft	3553.141409
3	hydrate	3464.200900
4	clean	3308.599579
5	sensitive	3059.395978
6	oily	2813.700901
7	smooth	2803.680878
8	moisturizer	2765.920943
9	acne	2561.834960
10	lips	2328.271191
11	gentle	2229.941992
12	texture	2187.194496
13	scent	2057.486804
14	pores	1728.229586
15	light	1691.717177
16	never	1690.712206
17	fresh	1609.892901
18	eye	1572.431135
19	buy	1570.365179

BIGRAMS

	term	score
0	acne prone	1835.260424
1	fine lines	1104.083796
2	exchange review	981.250564
3	skincare routine	939.291722
4	dry sensitive	907.557276
5	soft smooth	868.397290
6	holy grail	859.366650
7	soft hydrate	702.819032
8	lip balm	692.888077
9	double cleanse	672.851853
10	dry patches	664.406775
11	lips soft	633.278330
12	cleansing balm	633.204644
13	eye area	632.019108
14	soft clean	626.978582
15	dry lips	620.644283
16	easy apply	586.149358
17	drunk elephant	586.013853
18	fresh clean	584.113193
19	light weight	577.230506



	Concerns (from reviews)	Benefits (from reviews)
0	Acne	Hydrating
1	Aging	Brightening
2	Pores	Soothing
3	Redness	Plumping
4	Dark Spots	Firming
5	Texture	

	highlight_acne	highlight_aging	highlight_pores	highlight_redness	highlight_dark_spots	highlight_texture	highlight_hydrating	highlight_brightening	highlight_soothing	highlight_plumping	highlight_firming
6	0	0	1	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	1	0	0	0	0
15	0	0	0	0	0	0	1	0	0	0	0
16	0	0	0	0	0	0	1	0	0	0	0
18	0	0	0	0	0	0	1	0	0	0	0
25	0	0	0	0	0	0	1	0	0	0	0
26	0	0	0	0	0	0	1	0	0	0	0
29	1	0	0	0	0	0	0	0	0	0	0

Grouped into Concerns and Benefits & encoded as Binary Indicators

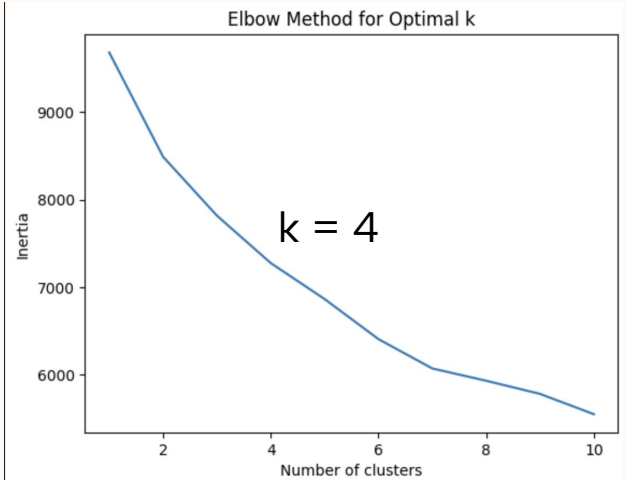
Clustering Model

Why We Cluster:

- Groups similar products together
- Find patterns in product features
- Improve recommendation quality

We Cluster Using:

- Ingredients, Skin Indicators…
- NOT Brand, Rating, Price…



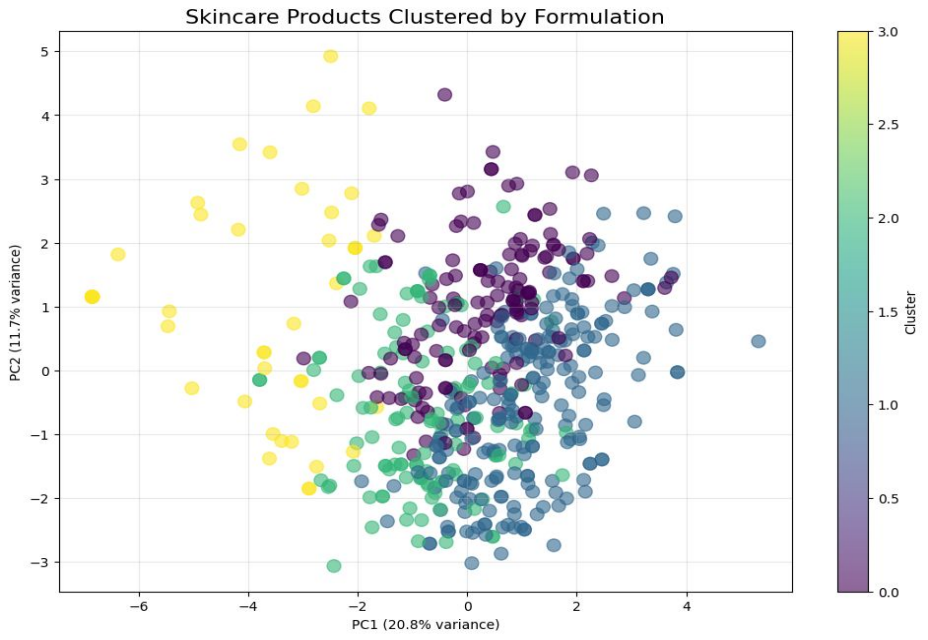
Our Clusters:

3 – Simple and Light Products

2 – Fragrance-Free / Sensitive Products

1 – Complex / Heavy Formulas

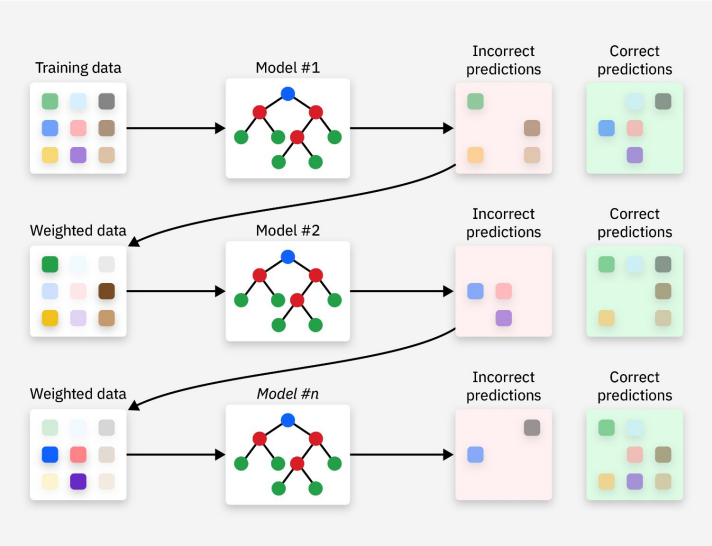
0 – Fragrances + Oil Heavy Moisturizers



Prediction Model

We Used a **Gradient Boosting Model**

- Sequential paths fix past mistakes
- Good for non-linear trends, mixing numerical and categorical data



Tuning the Model:

- Standardized numerical features
- Tested 50 unique models using different hyperparameter values
- Cross-Validated Models
- Kept the Model with Lowest Error

```
Number of trees used: 227

Final RMSE: 0.6279
Final R²: 0.6886
```

- Moderately Complex Model
- On average, ratings are within 0.63 stars of true rating
- Explains 69% of variation in data

Final Recommendation Model

- Takes information of the user's current routine and profile (skin type, skin concern)
- Clustering model: Finds the most common cluster
- If category or max price are provided by user, filters list in cluster.
- Uses the pre-trained XGboost model and scaler to predict a rating for each candidate product.
- Sorts the products by the highest predicted_rating and returns the top k results.

- No current routine?
- Clustering model bypassed and all products included in candidate pool.
- Same process as first function.

Example of input

```
example_Hayden = {
    "skin_dry": 1,
    "highlight_acne": 1,
    "highlight_redness": 1,
    "skin_oily": 1,
}
```

```
recommend_based_on_routine(
    user_profile=example_Hayden,
    k=10,
    category="Cleansers",
    max_price=20
)
```

Final Recommendation Model

	product_name	brand_name	secondary_category	price_usd	predicted_rating	cluster
1483	Clean Skin Gel Cleanser with Prebiotics	SEPHORA COLLECTION	Cleansers	12.00	4.777866	2
1205	Gentle Hydra-Gel Face Cleanser	NUDESTIX	Cleansers	19.00	4.771282	0
1680	Fulvic Acid Brightening Cleanser	The INKEY List	Cleansers	11.99	4.764445	2
1681	Mini Fulvic Acid Brightening Cleanser	The INKEY List	Cleansers	5.99	4.752027	2
1678	Salicylic Acid Acne + Pore Cleanser	The INKEY List	Cleansers	11.99	4.737176	1
1694	Mini Squalane Cleanser	The Ordinary	Cleansers	9.00	4.731672	1
830	Volcanic Pore Clearing Facial Cleanser with AH...	innisfree	Cleansers	14.00	4.721512	1
1493	Smoothing Cleanser	SEPHORA COLLECTION	Cleansers	12.00	4.720721	2
1691	Squalane Cleanser	The Ordinary	Cleansers	9.00	4.708479	1
828	Dewy Glow Jam Cleanser with Jeju Cherry Blossom	innisfree	Cleansers	13.00	4.683419	1

Products from clusters 1 and 2 dominate!

Products from cluster 2 seem to take higher top spots.

Limitations and Future Work

- **Limitations:**
  - XGBoost R^2 not yet satisfactory.
  - Clusters are not yet well defined.
  - Final model is not flexible.
- **Future Work:**
  - Improve text mining to extract more useful information out of reviews.
  - Try other models to predict rating.
  - PCA or other variable reduction methods.

Thank you!

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